

Data Preparation Session 9

Exercises

Exercise 9.1. (technical practice for regex)

- Let's create a variable called `text1` and populate it with the value "The dragon year is 2025."
- Create a variable called `my_pattern` and implement the required pattern for finding any digit in the variable `text1`. Use function `grepl` to verify if there is a digit in the string variable.
- Use function `gregexpr` to find all the positions in `text1` where there is a digit. Place the results in a variable called `string_position`.
- Create a variable called `my_pattern` and implement the required pattern for finding one digit and one uppercase alphanumeric character, in variable `text1`.
- Use function `regexpr` to find the position of the first space in `text1`. Place the results in a variable called `first_space`.
- Create a pattern that checks in `text1` if there is a lowercase character, followed by any character and then by a digit.
- Find the starting position of the above string. Place the results in a variable called `string_pos2`.
- Find the following pattern: one space followed by two lowercase letters and one more space. Use a function that returns the starting point of the found string and place its result in `string_pos3`.
- Using the `sub` function, replace the pattern found on the previous exercise by the string " is not ". Place the resulting string in `text2` variable.
- Find in `text2` the following pattern: Four digits starting at the end of the string. Use a function that returns the starting point of the found string and place its result in `string_pos4`.
- According to the position of the string found in the previous exercise, extract the first two digits starting at `string_pos4`.

Exercise 9.2. (quick, only a few steps)

- Load the `inpatient.tsv` data **with a not perfect separation** of DRG.

```
names <- c("DRG", "ProviderID", "Name", "Address", "City", "State", "ZIP", "Region", "Discharges",  
"AverageCharges", "AverageTotalPayments", "AverageMedicarePayments")  
types = 'ccccccccinnn'  
inpatient <- read_tsv(inpatient.tsv, col_names = names, skip=1, col_types = types)  
inpatient_separate <- separate(inpatient, DRG, c('DRGcode', 'DRGdescription'), 4)
```
- Trim the `DRGcode` field. Hint: `str_trim()`
- Remove the '–' from the beginning of the `DRGdescription` column.

Exercise 9.3 (retake of Exercise 2.3)

- Try to import the `employees_broken2.txt` file. Figure out the type.
- Check what part of the file is not data, but metadata.
- For the import, use `col_names = c("id", "name", "birthdate", "gender", "salary")`
- Use regular expressions to define `id`, `name`, `birthdate`, `gender` and `salary` (you can check what you found last week). Detect them and clean the data.

Exercise 9.4. (long exercise with a lot of small manipulations)

- a) Import the ForeignAid.tsv dataset. Pay attention to headings.
- b) Clean the country column. We have two countries, South Korea and North Korea.
- c) Clean the FundingAgency column. Try to find how many agencies are, and create a constant called "agency" that is the number of unique agencies.
- d) Transform the FundingAmount variable to numeric without creating NAs (you may need several manipulations: removing monetary symbols, commas, negative signs, etc. before changing the type).
- e) Create a frequency table for an adequate variable.